

# CTMC Models for Customer Journey Prediction

UCLA Stats 148 Capstone — Technical Reference

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## Contents

|          |                                                            |          |
|----------|------------------------------------------------------------|----------|
| <b>1</b> | <b>Problem Setup</b>                                       | <b>2</b> |
| <b>2</b> | <b>State-Space Choice</b>                                  | <b>2</b> |
| <b>3</b> | <b>Retained Baseline 1: Global CTMC</b>                    | <b>2</b> |
| 3.1      | Fitting . . . . .                                          | 2        |
| 3.2      | Prediction Target . . . . .                                | 2        |
| <b>4</b> | <b>Retained Baseline 2: Clustered CTMC</b>                 | <b>3</b> |
| <b>5</b> | <b>New Model 1: Timeout-Absorbing CTMC</b>                 | <b>3</b> |
| 5.1      | Construction . . . . .                                     | 3        |
| 5.2      | Prediction . . . . .                                       | 3        |
| <b>6</b> | <b>New Model 2: Piecewise Time-Varying Timeout CTMC</b>    | <b>3</b> |
| 6.1      | Motivation . . . . .                                       | 3        |
| 6.2      | Fitting . . . . .                                          | 4        |
| 6.3      | Prediction . . . . .                                       | 4        |
| <b>7</b> | <b>New Model 3: Empirical Semi-Markov Timeout Model</b>    | <b>4</b> |
| 7.1      | Motivation . . . . .                                       | 4        |
| 7.2      | Training Samples . . . . .                                 | 5        |
| 7.3      | Value Function . . . . .                                   | 5        |
| 7.4      | Is This a Markov Chain? . . . . .                          | 6        |
| 7.5      | Does It Use More History Than the Current Event? . . . . . | 6        |
| 7.6      | Comparison to Timeout-Absorbing CTMC . . . . .             | 6        |
| <b>8</b> | <b>Active Model Set</b>                                    | <b>7</b> |
| <b>9</b> | <b>Commands</b>                                            | <b>7</b> |

# 1 Problem Setup

Each customer journey is a timestamped sequence of events. The project label is binary:

$$y = \begin{cases} 1 & \text{journey reaches } \texttt{order\_shipped}, \texttt{ed\_id}=28, \\ 0 & \text{journey fails by taking no customer action for 60 days.} \end{cases}$$

This is not exactly the same target as the original CTMC experiments. The first CTMC models estimated

$$P(T_{\text{success}} \leq 60 \text{ days} \mid X(0) = s),$$

the probability of hitting the success state within a fixed horizon. The classification target is better described as

$$P(T_{\text{success}} < T_{\text{inactivity-failure}} \mid X(0) = s),$$

where the failure time occurs when no customer action happens for 60 days.

## 2 State-Space Choice

The file `data/Event Definitions.csv` mixes customer actions with company or system events. For the inactivity definition, company/system events should not reset the 60-day clock. Therefore the active CTMC state space uses customer-initiated actions only:

$$\{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 18, 19, 22, 23, 25\}.$$

The success event `order_shipped` (`ed_id=28`) is retained as a terminal outcome state, not as a customer action. The timeout-aware models also add an artificial absorbing failure state -1.

This design intentionally excludes states such as approval, decline, pending, catalog mail, downpayment received/cleared, fraud review, account activation, and order fulfillment events from the behavioral CTMC state space.

## 3 Retained Baseline 1: Global CTMC

### 3.1 Fitting

For consecutive customer-action transitions  $(i, j, \Delta t)$ , define

$$N_{ij} = \#\{(i, j, \cdot)\}, \quad T_i = \sum_{(i, j, \Delta t)} \Delta t.$$

The maximum-likelihood CTMC generator is

$$\hat{q}_{ij} = \frac{N_{ij}}{T_i}, \quad i \neq j, \quad \hat{q}_{ii} = -\sum_{j \neq i} \hat{q}_{ij}.$$

### 3.2 Prediction Target

The global baseline still predicts the old target:

$$\hat{p}_{\text{global}}(s) = [\exp(Q^* \tau)]_{s, 28}, \quad \tau = 60 \cdot 86400.$$

Here  $Q^*$  makes state 28 absorbing. This is useful as a baseline but it does not explicitly model inactivity failure.

## 4 Retained Baseline 2: Clustered CTMC

The clustered CTMC groups journeys by prefix behavior and fits one global CTMC per cluster:

$$Q^{(k)} = \text{GlobalCTMC.fit}(\mathcal{D}_k).$$

The implementation supports two clustering modes:

| Mode     | Switch                                    | Notes                                                              |
|----------|-------------------------------------------|--------------------------------------------------------------------|
| KMeans   | default                                   | Original baseline; clusters action proportions and state features. |
| Spectral | <code>use_spectral_clustering=True</code> | Nearest-neighbor spectral fit plus KNN assignment.                 |

The clustered CTMC also predicts the old success-hit target unless paired with one of the timeout-aware models below.

## 5 New Model 1: Timeout-Absorbing CTMC

### 5.1 Construction

The timeout transition table modifies the training data:

- Customer-action transitions are kept.
- Transitions into `order_shipped=28` are kept as success.
- Each incomplete journey receives one artificial terminal transition from its last customer-action state to failure state  $f = -1$  after 60 days.

A CTMC generator is then fit on this augmented state space. Both success and failure are absorbing.

### 5.2 Prediction

For transient states  $\mathcal{T} = \mathcal{S} \setminus \{28, -1\}$ , the probability of eventually absorbing in success before failure solves

$$Q_{\mathcal{T}\mathcal{T}}u = -Q_{\mathcal{T},28}.$$

The model returns  $u_s$  for the current state  $s$ . This directly targets

$$P(T_{\text{success}} < T_{\text{failure}} \mid X(0) = s).$$

## 6 New Model 2: Piecewise Time-Varying Timeout CTMC

### 6.1 Motivation

The timeout-absorbing CTMC assumes one fixed generator matrix  $Q$  for the whole journey. This can average together different behavioral regimes. In practice, customer behavior may

change with journey age: very early journeys are often application or browsing heavy, while older journeys are more likely to decay toward inactivity.

The piecewise time-varying model approximates a non-homogeneous CTMC by using one generator matrix per journey-age window:

$$Q(t) = Q_b \quad \text{when } t \in I_b.$$

The default bins are

$$[0, 1\text{d}), [1, 7\text{d}), [7, 30\text{d}), [30, 60\text{d}), [60\text{d}, \infty).$$

## 6.2 Fitting

For each timeout-aware transition  $(i, j, \Delta t)$ , the model tracks the cumulative journey age at the start and end of the dwell interval. Exposure time in state  $i$  is split across any bins crossed by that interval. The jump count  $i \rightarrow j$  is assigned to the bin where the jump occurs. Thus, in bin  $b$ ,

$$\hat{q}_{ij}^{(b)} = \frac{N_b(i, j)}{T_b(i)}, \quad i \neq j,$$

where  $T_b(i)$  is the total exposure time spent in state  $i$  during age window  $b$ , and  $N_b(i, j)$  is the number of observed jumps from  $i$  to  $j$  whose jump time falls in that same window. The diagonal is again set by

$$\hat{q}_{ii}^{(b)} = -\sum_{j \neq i} \hat{q}_{ij}^{(b)}.$$

Rows with too little bin-specific data fall back to the corresponding row of the global timeout-aware generator. The success state 28 and timeout state -1 remain absorbing in every bin.

## 6.3 Prediction

Prediction uses true piecewise propagation. Starting from current journey age  $t$  and current state  $s$ , the model advances the probability row vector across all bin boundaries crossed by the 60-day horizon:

$$p(t + h) = p(t) \exp(Q_{b_1} \Delta t_1) \exp(Q_{b_2} \Delta t_2) \cdots \exp(Q_{b_m} \Delta t_m).$$

The returned score is the probability mass in `order_shipped` after this propagation. This keeps the CTMC interpretation but lets rates vary with journey age.

# 7 New Model 3: Empirical Semi-Markov Timeout Model

## 7.1 Motivation

The timeout-absorbing CTMC is still a continuous-time Markov chain. Once a journey is in state  $i$ , it assumes the waiting time until the next transition is exponential with rate

$$\lambda_i = -q_{ii}.$$

This is mathematically convenient, but it imposes the memoryless property: conditional on still being in state  $i$ , the remaining waiting-time distribution is the same no matter how long the journey has already waited in that state.

The empirical Semi-Markov timeout model is a less parametric state-based alternative. It keeps the same customer-action state space and the same absorbing outcomes, but it does not estimate a generator matrix  $Q$  and does not assume exponential holding times. Instead, it uses the empirical distribution of observed next events and observed waits after each state.

## 7.2 Training Samples

For each observed transition out of state  $i$ , the model stores a sample

$$(j_m, \Delta t_m),$$

where  $j_m$  is the next observed modeling state and  $\Delta t_m$  is the elapsed time until that next state. These samples come from the same timeout-aware transition table used by the timeout-absorbing CTMC. Therefore:

- transitions into `order_shipped=28` are success samples;
- transitions into the artificial state `-1` are inactivity-failure samples;
- waits longer than 60 days are treated as timeout failures even if an eventual later action exists in the raw data.

For example, from a state such as `view_cart`, the model may have many historical samples of the form

$$\text{view\_cart} \rightarrow \{\text{begin\_checkout}, \text{add\_to\_cart}, \text{order\_shipped}, -1\},$$

each with its own observed waiting time. The model uses these samples directly rather than compressing them into one exponential rate.

## 7.3 Value Function

The model estimates

$$V(i) = P(\text{eventually ship before inactivity failure} \mid X = i).$$

The absorbing boundary values are fixed:

$$V(28) = 1, \quad V(-1) = 0.$$

For a nonterminal state  $i$ , let

$$\mathcal{D}_i = \{(j_m, \Delta t_m) : m = 1, \dots, M_i\}$$

be the empirical outgoing samples from state  $i$ . The value update is

$$V(i) = \frac{1}{M_i} \sum_{m=1}^{M_i} \begin{cases} 1, & j_m = 28 \text{ and } \Delta t_m \leq 60 \text{ days,} \\ 0, & j_m = -1 \text{ or } \Delta t_m > 60 \text{ days,} \\ V(j_m), & \text{otherwise.} \end{cases}$$

The implementation initializes the nonterminal values and repeatedly applies this update until the values stabilize. This is analogous to solving an absorbing Markov reward/value equation, except that the transition mechanism is estimated from empirical next-step samples and empirical holding times rather than from a CTMC generator.

## 7.4 Is This a Markov Chain?

Strictly speaking, this model is not a CTMC, because it does not use exponential holding times and does not define a generator matrix  $Q$ . It is also not a discrete-time Markov chain in the simplest textbook sense, because each transition sample includes a duration  $\Delta t_m$  and the timeout rule depends on that duration.

It is best described as a *state-based empirical semi-Markov model*. It is “Markov” in the sense that its prediction uses the current state  $i$  as the state variable, and the future is summarized through empirical outcomes observed after visits to that state. It is “semi-Markov” because the next state and the waiting time are modeled jointly through samples  $(j_m, \Delta t_m)$  rather than through memoryless exponential holding times.

## 7.5 Does It Use More History Than the Current Event?

In the current implementation, no. At prediction time the model only receives the current modeling state, i.e. the most recent customer-action event. It does not condition on the full prefix, the number of prior actions, the elapsed age of the journey, recency, user-specific covariates, or the previous sequence of states.

Historical data are used during fitting to estimate the empirical outgoing sample distribution for each state. But once the model is fit, two open journeys whose current state is `view_cart` receive the same Semi-Markov score, even if one arrived there after one click and another arrived there after a long complex path. This makes the model interpretable and strong as a state-based benchmark, but it is not personalized. Models such as the truncated-snapshot clustered CTMC are designed to use richer prefix history.

## 7.6 Comparison to Timeout-Absorbing CTMC

The timeout-absorbing CTMC asks:

Given fitted transition rates  $q_{ij}$ , what is  $P(T_{\text{success}} < T_{\text{failure}})$ ?

The Semi-Markov model asks:

Given the empirical next-step outcomes after this state, what fraction of future paths lead to success before timeout?

The tradeoff is therefore:

| Model                         | Waiting-time assumption  | Extra history? |
|-------------------------------|--------------------------|----------------|
| Timeout-Absorbing CTMC        | exponential / memoryless | no             |
| Empirical Semi-Markov Timeout | empirical waits by state | no             |

The Semi-Markov model can be more realistic when customer waits are not well described by an exponential distribution. However, because it uses empirical samples state-by-state, rare states can be noisier than in the smoother CTMC rate model.

## 8 Active Model Set

The active CTMC comparison contains the retained baselines plus several timeout-aware extensions:

| Model                           | Target                              | Status                      |
|---------------------------------|-------------------------------------|-----------------------------|
| GlobalCTMC                      | hit success within 60 days          | retained old baseline       |
| ClusteredCTMC                   | hit success within 60 days          | retained old baseline       |
| TimeoutAbsorbingCTMC            | success before inactivity failure   | new primary CTMC target     |
| PiecewiseTimeVaryingTimeoutCTMC | time-varying success before timeout | non-homogeneous CTMC target |
| SemiMarkovTimeoutModel          | success before inactivity failure   | new non-exponential target  |

Older experimental CTMC variants such as neural-rate CTMC, XGBoost-rate CTMC, legacy time-stratified CTMC, Laplace-smoothed CTMC, and isotonic CTMC calibration are not part of the active CTMC comparison.

## 9 Commands

Run the retained CTMC comparison:

```
python -m src.models.ctmc_pipeline all --no-cache
```

Run only spectral clustered CTMC:

```
python -m src.models.ctmc_pipeline clustered  
--use-spectral-clustering --no-cache
```

Run the piecewise time-varying timeout CTMC:

```
python -m src.models.ctmc_pipeline temporal  
--test-events data/open_journeys2.csv  
--sample data/open_journeys2_flattened_all0.csv  
--output-dir results/submissions/open_journeys2
```